BoltPe Financial Planner

Frontend Validation & Error Handling Guide

Based on backend code: financial_planner_go (184)

Purpose: provide the frontend team with the exact validation rules, backend messages, and UI handling guidance required to prevent invalid Financial Planner requests before API submission.

1. Purpose and Scope

- Primary APIs covered: POST /api/financial-planner/report-json and POST /api/financial-planner/report-pdf-direct.
- Validation errors return HTTP 400 with code VAL_001 and success=false.
- Frontend should validate locally using the same conditions, but backend validation remains the final source of truth.
- The document also lists non-validation errors that frontend cannot prevent and should display or handle generically.

2. Standard Error Response

```
{
  "code": "VAL_001",
  "message": "<validation message>",
  "success": false
}
```

Property	Meaning	Frontend handling
code	Machine-readable error code. Validation uses VAL_001.	Use for routing/general handling; do not replace backend message.
message	Human-readable backend error.	Show near field/section when mapped; otherwise show form-level error.
success	false for error responses.	Treat request as failed.

3. Goal Selection Mode - Source of Truth

goal_selection_mode	Allowed selected_goals_json types	Family / Insurance behavior
instant	Only instant	Strict Instant rules: family details not allowed; insurance not allowed.
short	Only short	Normal family and insurance rules.
long	Only long	Normal family and insurance rules.
create_your_own	Any mix of instant, short, long	Normal family and insurance rules; Instant child goals do NOT activate Instant-only restrictions.

Field / rule	Trigger	Exact backend message	Frontend handling
goal_selection_mode	Required.	goal_selection_mode is required and must be one of: instant, short, long, create_your_own	Require a top-level selection before submit.
goal_selection_mode	Value outside supported list.	goal_selection_mode '<value>' is invalid; supported values are: instant, short, long, create_your_own	Use enum/dropdown only.
selected_goals_json	Empty array.	selected_goals_json must contain at least one goal	Require at least one goal.
goal type vs selection mode	For direct instant/short/long, every child type must match mode.	selected_goals_json[i].type '<type>' is not allowed when goal_selection_mode is '<mode>'; only '<mode>' goals are allowed. Use goal_selection_mode 'create_your_own' to select multiple goal types	Disable other categories when direct mode is selected.

4. Customer / Profile Validation

Field	Trigger	Exact backend message	Frontend rule
full_name	Length <3 or >100	full_name must be between 3 and 100 characters	Required; 3-100 chars.
full_name	Invalid characters	full_name may contain only letters, spaces, apostrophes, dots, and hyphens	Restrict characters / inline error.
contact	Not Indian 10-digit mobile starting 6-9	contact must be valid 10 digit Indian mobile number	Regex: ^[6-9][0-9]{9}$.
email	Invalid email syntax	email must be valid	Use email input + syntax validation.
dob	Invalid format	dob must be in YYYY-MM-DD format	Date input; send YYYY-MM-DD.
dob	Before 1900 or future date	dob year must be between 1900 and current year and cannot be future date	Set min/max date.
dob	Calculated age <18 or >100	customer age calculated from dob must be between 18 and 100 years	Calculate age before submit.
city	Empty	city is required	Required.
city	Length <2 or >100	city must be between 2 and 100 characters	2-100 chars.
risk_appetite	Not low/medium/high	risk_appetite must be one of: low, medium, high	Enum control.
investment_experience	Not Beginner/Intermediate/Advanced	investment_experience must be one of: Beginner, Intermediate, Advanced	Enum control. Backend normalizes letter case for known values.

5. Family / Marital Validation

Important: Instant top-level mode is a special case. create_your_own is not treated as Instant even when it contains only Instant child goals.

Field / scenario	Trigger	Exact backend message	Frontend rule
Instant mode - Family Details	Any family detail supplied in Instant mode (if applicable)	Family details are not applicable for instant goals. marital_status and child details must be empty; spouse_age, spouse_income, children_count, and dependents must be empty or 0	Hide/clear family section for Instant mode.
dependents	Explorer (Create Your Own) and omitted	dependents is required when selected goals contain multiple goal types (Explorer flow)	Mandatory for Explorer flow.
dependents	Not numeric	dependents must be a whole number	Integer input only.
dependents	Negative value	dependents cannot be negative	Minimum value 0.
marital_status	Empty	marital_status is required	Mandatory field.
marital_status	Invalid value	marital_status must be single or married	Dropdown with only single and married.
spouse_dob	Invalid format	dob must be in YYYY-MM-DD format	Date input; send YYYY-MM-DD.
spouse_income	single and non-zero	spouse_income must be 0 when marital_status is single	Reset to 0/blank.
children_count	single and non-zero	children_count must be 0 when marital_status is single	Set 0.
spouse_age	married and omitted	spouse_age is required when marital_status is married	Mandatory.
spouse_income	married and omitted	spouse_income is required when marital_status is married	Mandatory.
children_count	married and omitted	children_count is required when marital_status is married	Mandatory (0 allowed).
spouse_age	Not numeric	spouse_age must be a valid number	Integer only.
spouse_age	Less than 18	spouse_age must be at least 18 when marital_status is married	Minimum 18.
spouse_income	Negative	spouse_income cannot be negative	Minimum 0.
children_count	Not numeric	children_count must be a valid number	Integer only.
children_count	Negative	children_count cannot be negative	Minimum 0.
child details	children_count=0 but child details supplied	child details cannot be provided when children_count is 0	Hide child section.
child_N	Index exceeds children_count	child_N details cannot be provided when children_count is X	Render only required child forms.
child_N	Duplicate index	child_N details are duplicated	Prevent duplicate indexes.
child_N_name	Invalid name or >100 chars	child_N_name must be a valid name up to 100 characters	Max 100 chars.
child_N_name	Name missing but age provided	child_N_name is required when child_N_age is provided	Require both fields.
child_N_age	<0 or >25	child_N_age must be between 0 and 25	Allow 0-25.
Customer & Child Age	Gap <18 years	child_N_age is invalid; parent age must be at least 18 years greater than child age	Validate 18-year gap.
Spouse & Child Age	Gap <18 years	child_N_age is invalid; spouse age must be at least 18 years greater than child age	Validate 18-year gap.
child_dob	Invalid format	dob must be in YYYY-MM-DD format	Date input; send YYYY-MM-DD.
Child Age	>= Customer age	child_N_age must be less than customer age	Child younger than customer.
Child Age	>= Spouse age	child_N_age must be less than spouse age	Child younger than spouse.
Multiple Children	Duplicate names	duplicate child names are not allowed	Prevent duplicates.

Multiple Children	Duplicate records	duplicate child details are not allowed	Prevent duplicates.
children_count	Less than child records	child details exceed children_count	Limit child records.
children_count	Greater than supported maximum	children_count exceeds maximum supported limit	Restrict maximum.

5.1. Age Relationship Validations

Validation	Backend Message	Frontend Rule
Customer Age < 18	customer age must be at least 18 years	Minimum customer age is 18.
Married & Spouse Age < 18	spouse_age must be at least 18 when marital_status is married	Minimum spouse age is 18.
Customer Age − Child Age < 18	child_N_age is invalid; parent age must be at least 18 years greater than child age	18-year gap.
Spouse Age − Child Age < 18	child_N_age is invalid; spouse age must be at least 18 years greater than child age	18-year gap.
Child Age ≥ Customer Age	child_N_age must be less than customer age	Child younger than customer.
Child Age ≥ Spouse Age	child_N_age must be less than spouse age	Child younger than spouse.

6. Generic Numeric Validation

The following fields are validated as non-negative numeric values when provided. Whole-number fields cannot contain decimals. Decimal-capable fields allow a maximum of 2 decimal places.

Category	Fields
Whole numbers	dependents, children_count, spouse_age, education_1_target_age, marriage_1_target_age
Decimal-capable	monthly_income, monthly_expenses, monthly_savings, spouse_income, all insurance cover fields, all inflation/return assumption fields

Condition	Backend message pattern	Frontend handling
More than 2 non-zero decimal places	<field> can have maximum 2 decimal places	Limit/round to 2 decimals.
Not numeric	<field> must be numeric	Numeric input only.
Negative	<field> cannot be negative	Min 0.
Whole-number field has decimal	<field> must be a whole number	Integer only.

7. Cashflow Validation

Field	Trigger	Exact backend message	Frontend rule
monthly_income / monthly_expenses / monthly_savings	Any one omitted	monthly_income, monthly_expenses, and monthly_savings are required	All three required.
monthly_income	<= 0	monthly_income must be greater than 0	Must be >0.
monthly_expenses	> monthly_income	monthly_expenses cannot be greater than monthly_income	Cross-field validation.
monthly_savings	> monthly_income	monthly_savings cannot be greater than monthly_income	Cross-field validation.
expenses + savings	> monthly_income	monthly_expenses plus monthly_savings cannot be greater than monthly_income	Show form-level cashflow error.

8. Insurance Validation

Direct Instant mode: all insurance selections must represent false/no/0 or be omitted, and every cover must be 0/empty.

```
insurance details are not applicable for instant goals; insurance selections must be no/false/0 and cover amounts must be 0
```

Short / Long / Create Your Own: normal insurance validation applies.

Rule	Exact backend message pattern	Frontend handling
Flag value invalid	<flag> must be one of: yes, no, true, false, 1, 0	Use boolean/toggle; accepted text variants include yes/no, true/false, y/n, 1/0.
Cover non-numeric	<cover> must be numeric	Numeric input.
Cover >0 but flag omitted	<flag> is required when <cover> is greater than 0	Selecting/entering cover should set flag.
Flag selected=true but cover <=0	<cover> must be greater than 0 when <flag> is selected	Require positive cover.
Flag false but cover !=0	<cover> must be 0 when <flag> is not selected	Clear cover on toggle off.

Insurance pairs: existing_life_term + existing_life_term_cover; existing_health + existing_health_cover; existing_critical + existing_critical_cover; existing_personal + existing_personal_cover.

9. Assumption Validation

Fields	Allowed range	Notes / backend pattern
education_1_inflation, marriage_1_inflation, retirement_inflation, wealth_inflation, generic_goal_inflation	0-20	Optional. Error: <field> must be between 0 and 20.
All lumpsum/SIP return fields + retirement_pre_return + retirement_post_return	0-50	Optional. Error: <field> must be between 0 and 50.
education_1_target_age, marriage_1_target_age	1-100 when non-zero	Optional; empty/0 skipped. Must be whole number. Error: <field> must be between 1 and 100.

10. Goal Master and Goal Item Validation

Type	Allowed keys / labels	Timeline range
instant	iphone - iPhone; playstation - PlayStation; bike - Bike; tv - TV; fridge - Fridge; washing_machine - Washing Machine; vacation - Vacation Plan; others - Others	6-12 Months
short	self_development - Self Development / Higher Education; home_purchase - Home Purchase / Home Upgrade; car_purchase - Car Purchase; own_marriage - Own Marriage Planning	1-5 Years
long	retirement - Retirement Planning; wealth - Wealth Creation	20-25 Years

Validation	Backend message pattern
type missing	selected_goals_json[i].type is required
invalid type	selected_goals_json[i].type '<type>' is invalid; supported types are: instant, short, long
key missing	selected_goals_json[i].key is required for type '<type>'
unknown key for type	selected_goals_json[i].key '<key>' is invalid for type '<type>'; supported <type> goal keys are: ...
key belongs to another type	selected_goals_json[i].key '<key>' is invalid for type '<type>'; the goal '<key>' belongs to type '<actual>'. Select type '<actual>'. Supported <type> goal keys are: ...
duplicate key	selected_goals_json[i].key '<key>' must be unique
label missing	selected_goals_json[i].label is required for <type> goal key '<key>'
label does not exactly match Goal Master	selected_goals_json[i].label '<label>' is invalid for <type> goal key '<key>'; expected label is '<expected>'
amount <=0	selected_goals_json[i].amount must be greater than 0
others without customName	selected_goals_json[i].customName is required for instant goal key 'others'

11. Goal Timeline Validation

The display timeline must agree with the numeric years/months fields. “Within ” is accepted by the parser, but unit and range rules still apply.

Validation	Backend message pattern
Timeline missing	selected_goals_json[i].timeline is required for <type> goal key '<key>'
Instant invalid format	selected_goals_json[i].timeline must be in month format, for example '12 Months', for instant goal key '<key>'
Short/Long invalid format	selected_goals_json[i].timeline must be in year format, for example '<min> Years', for <type> goal key '<key>'
Invalid numeric text	selected_goals_json[i].timeline contains an invalid numeric value
Instant wrong unit	selected_goals_json[i].timeline must use Months for instant goal key '<key>'
Instant outside range	selected_goals_json[i].timeline must be between 6 and 12 Months for instant goal key '<key>'

Instant mismatch with months	selected_goals_json[i].timeline '<timeline>' does not match months=<n> for instant goal key '<key>'
Short/Long wrong unit	selected_goals_json[i].timeline must use Years for <type> goal key '<key>'
Short outside range	selected_goals_json[i].timeline must be between 1 and 5 Years for short goal key '<key>'
Long outside range	selected_goals_json[i].timeline must be between 20 and 25 Years for long goal key '<key>'
Year mismatch	selected_goals_json[i].timeline '<timeline>' does not match years=<n> for <type> goal key '<key>'
Instant months outside range	selected_goals_json[i].months must be between 6 and 12 for instant goal key '<key>'
Instant years non-zero	selected_goals_json[i].years must not be provided for instant goal key '<key>'
Short years outside range	selected_goals_json[i].years must be between 1 and 5 for short goal key '<key>'
Short months non-zero	selected_goals_json[i].months must not be provided for short goal key '<key>'
Long years outside range	selected_goals_json[i].years must be between 20 and 25 for long goal key '<key>'
Long months non-zero	selected_goals_json[i].months must not be provided for long goal key '<key>'

12. Request Parsing / Payload Errors

Scenario	Backend response/message	Frontend handling
Body unreadable	VAL_001 - Unable to read request payload	Generic form error; log client-side diagnostics.
Empty or invalid JSON	VAL_001 - Invalid financial planner request payload	Should never occur with normal JSON serialization.
selected_goals_json missing during custom unmarshal	selected_goals_json is required	Always send array.
selected_goals_json malformed stringified array	selected_goals_json must be a valid stringified JSON array: <error>	Prefer normal JSON array.
selected_goals_json neither array nor valid stringified array	selected_goals_json must be a JSON array or stringified JSON array: <error>	Prefer normal JSON array.
years/months invalid type/value	years/months must be a whole number or an empty string	Send integer, numeric string, empty string, null, or omit.
child_N_name wrong type	child_N_name must be a string	Send string only.
child_N_age invalid	child_N_age must be a whole number or an empty string	Send integer/numeric string/empty/null.

13. Recommended Frontend Validation Flow

1. Require goal_selection_mode first.

2. When mode is instant/short/long, restrict goal picker to that type only. When mode is create_your_own, allow all three types and multiple selections.

3. For direct Instant mode, hide and clear family + insurance sections before constructing the request.

4. For short/long/create_your_own, show marital/family logic and insurance logic.

5. Validate customer profile and cashflow fields.

6. Validate selected goals against the Goal Master, unique keys, exact label, amount, type, timeline, years/months, and customName for others.

7. Validate optional assumptions only when values are entered.

8. Submit only after local validation passes; still display backend VAL_001 messages because backend is authoritative.

14. Frontend Field Visibility / Requirement Matrix

Field / section	Instant	Short	Long	Create Your Own
goal_selection_mode	Required	Required	Required	Required
selected goals	Instant only	Short only	Long only	Any instant/short/long mix
marital_status	Must be empty	Required: single/married	Required: single/married	Required: single/married
spouse fields	Empty/0	Conditional	Conditional	Conditional
children_count	Empty/0	Required, 0 allowed	Required, 0 allowed	Required, 0 allowed
dependents	Empty/0	Optional (defaults to 0)	Optional (defaults to 0)	Required
child details	Not allowed	Optional if count >0	Optional if count >0	Optional if count >0
insurance	Not allowed	Allowed + validated	Allowed + validated	Allowed + validated

15. Non-Validation Error Codes for Frontend

These errors generally cannot be prevented with form validation. Frontend should show a safe error message, retain user input, and allow retry where appropriate.

Code	Meaning / module	Typical handling
FP_CALC_001	Planning/calculation engine failure	Show generic calculation failure; allow retry.
FP_CALC_INVALID_RESPONSE	Invalid calculation response	Generic service error; allow retry/support escalation.
FP_RESPONSE_ERROR	Response serialization error	Generic service error.
FP_STORAGE_ERROR	Database/storage failure	Inform user request could not be stored; retry.
FP_REPORT_001	PDF generation failure	JSON may exist; show PDF generation failure/retry download.
FP_STORAGE_001	S3/object-storage failure	Show report storage/download issue.
FP_EMAIL_001	Email delivery failure	Report may still exist; show email-specific failure.
FP_LIST_500	Report list fetch failure	Show list loading error/retry.
FP_404	Generated report not found	Show report unavailable/not found.
VAL_001	Input validation / request format failure	Map to fields/sections; prevent when possible.

16. Implementation Notes for Frontend Team

- Do not derive validation only from selected_goals_json. goal_selection_mode controls the top-level flow.
- For create_your_own, never apply Instant-only family/insurance restrictions just because all selected child goals happen to be instant.
- Use exact Goal Master key + label pairs. Backend label comparison is exact after trimming in the main validator.
- Clear hidden conditional values before submit. Hidden stale values can trigger backend validation.
- For dynamic child fields, use child_1_name, child_1_age, child_2_name, child_2_age, etc. Do not submit indices greater than children_count.
- Prefer selected_goals_json as a normal JSON array; backend also supports a stringified array only for legacy compatibility.
- Backend validation order means the first failing rule is usually returned. Frontend may show multiple local errors at once, but should preserve the backend message if API rejects the payload.

17. Example: Direct Instant vs Create Your Own

Direct Instant - family and insurance must be empty/zero


```
{
  "goal_selection_mode": "instant",
  "marital_status": "",
  "spouse_age": 0,
  "spouse_income": 0,
  "children_count": 0,
  "dependents": 0,
  "existing_life_term": false,
  "existing_life_term_cover": 0,
  "selected_goals_json": [{"key":"iphone","label":"iPhone","type":"instant","timeline":"12
Months","amount":90000,"years":0,"months":12}]
}
```


Create Your Own - family and insurance allowed, even with Instant child goals


```
{
  "goal_selection_mode": "create_your_own",
  "marital_status": "married",
  "spouse_age": 30,
  "children_count": 1,
  "dependents": 1,
  "existing_life_term": true,
  "existing_life_term_cover": 10000000,
  "selected_goals_json": [{"key":"iphone","label":"iPhone","type":"instant","timeline":"12
```

Months","amount":90000,"years":0,"months":12}]
}